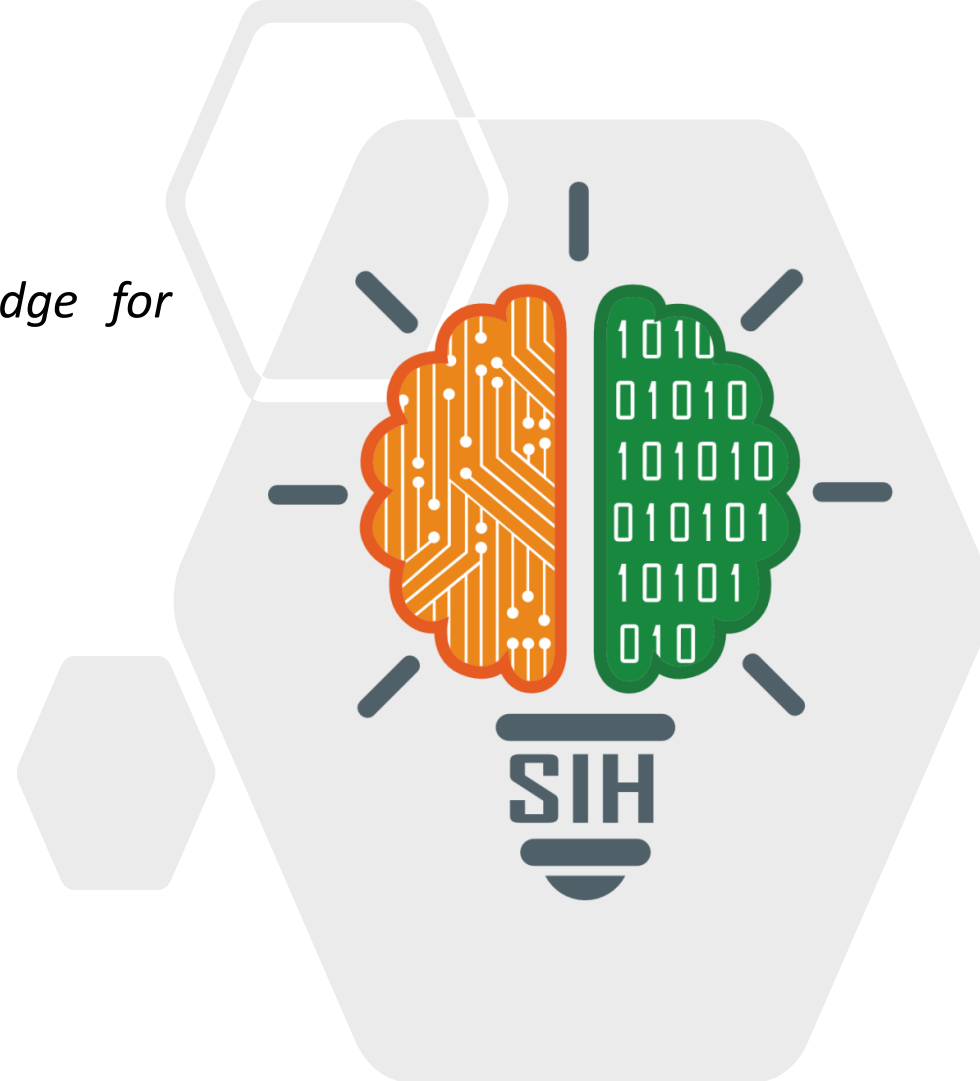


SMART INDIA HACKATHON 2024



- **Problem Statement ID – 1757**
- **Problem Statement Title - *India Post A Bridge for Indian Diaspora to access things Indian***
- **Theme - *Transportation & Logistics***
- **PS Category - *Software***
- **Team ID - 4326**
- **Team Name - *The Deciders***

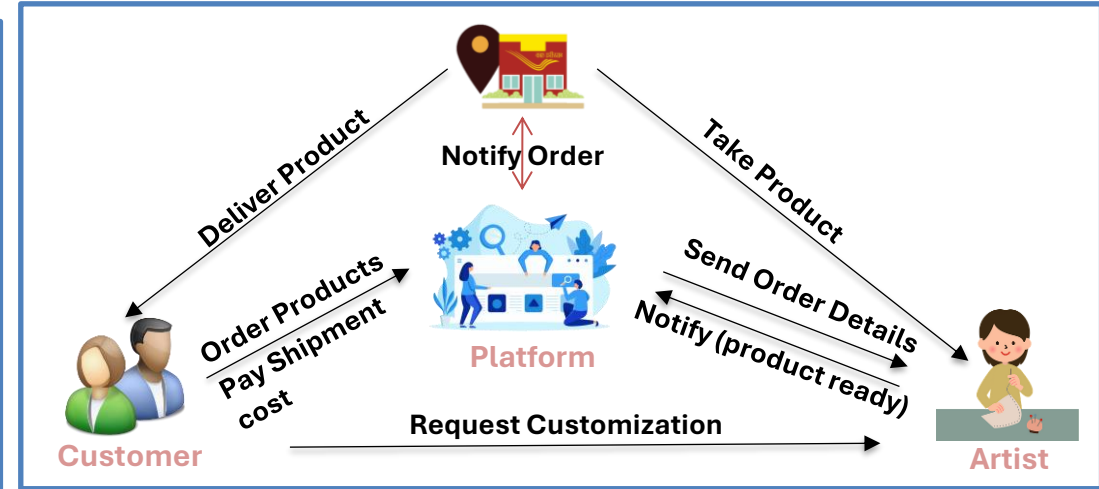


Proposed Solution (Describe your Idea/Solution/Prototype)

Idea/Solution :-

A web application that provides access of authentic Indian products from **MSMEs** to the **global Indian diaspora & international buyers** at a reasonable price, meanwhile promoting and preserving traditional craftsmanship and local artisans. The platform will:

- Analyze data of around **73% diaspora people & foreigners** buy Indian products online, with the help of **AI** and **ML** algorithms to provide results and offers based on their needs and preferences, enhancing user satisfaction.
- Provide exclusive collection during **festivals** and **seasonal** events (as demand rises by **40%**). User will also be provided with the option to order **customizable product** according to their own choice.
- Provide “**DoP Certified**” badge to trusted sellers after proper checks.
- Use **DoP’s (Department of post’s)** global network across **219** countries for affordable & reliable shipping (**30-50%** cheaper than private courier services).
- Ensure **secure payment** processing and product verification to guarantee authenticity and quality, addressing concerns of over **60%** of Indian diaspora buyers who demand trusted & certified sources.



Unique Selling Proposition (USP) :-

- Users can **customize products** according to their preferences, increasing personalized experience.
- Product offering according to different **Indian Festivals**
- **Chat** feature allows sellers and artisans to **communicate** directly ensuring **transparency**.
- **Multicurrency Support** for international Customer
- **Smart Contracts** will be made once product is finalized to ensure security & authenticity at both ends.

Technology Stack :-

Frontend Development :-

React JS, Tailwind CSS, Next JS

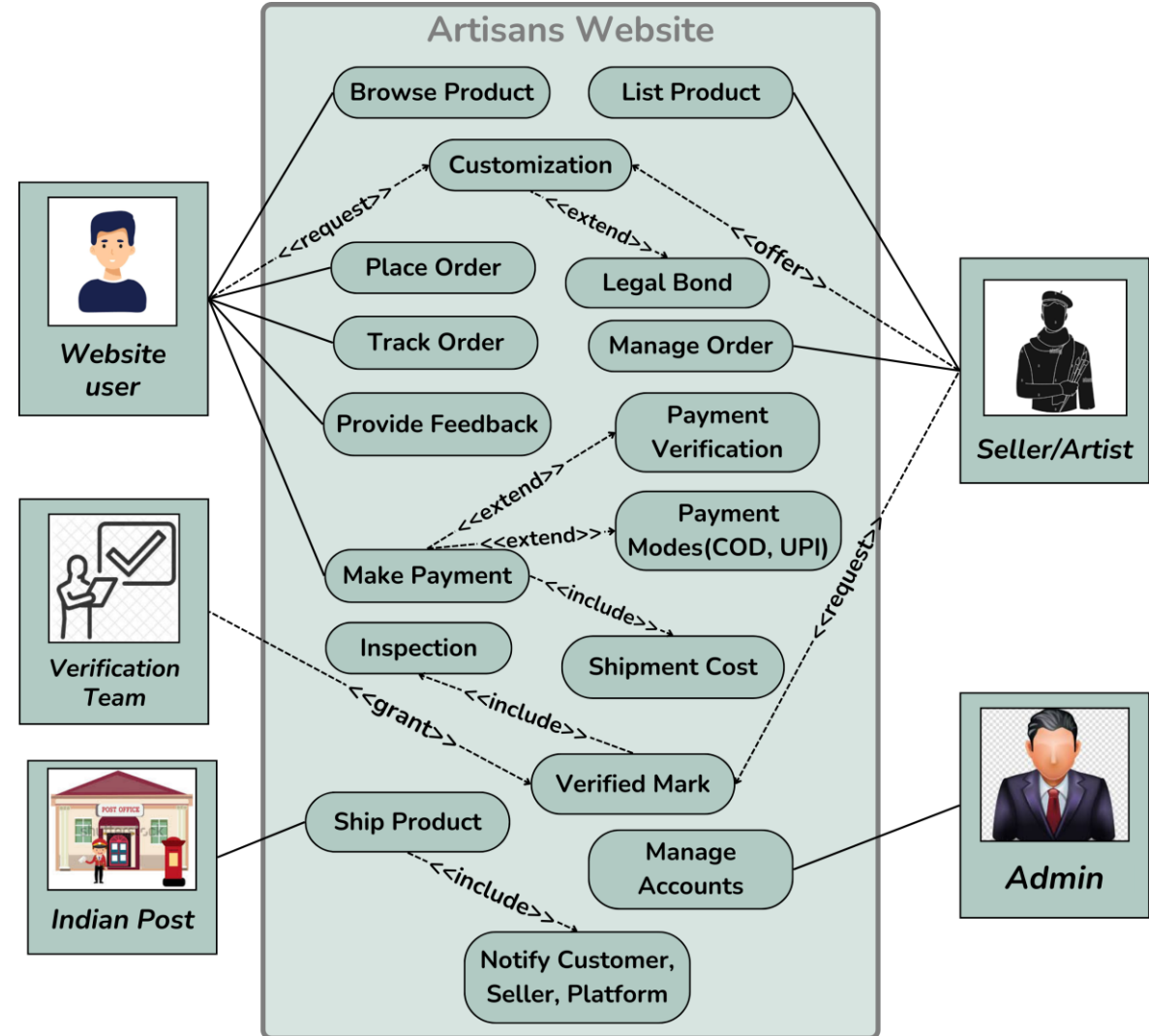
Backend Development :-

Node.js, Express.js, MongoDB, REST API

AI & Data Analysis :-

- **Apache Hadoop** to analyze user data, track sales trends etc. for personalized recommendations.
- **Google BigQuery and NLP(Natural Language Processing)** for predictive analysis enabling data-driven decisions and demand forecasting.
- **Power BI** to extract market insights for optimized product offerings, categorization and improve overall user engagement.
- **Blockchain Ledger** be used to record product's journey at each step (creation, quality checks, shipping etc.)

Use Case Diagram



Feasibility :-

Technical Feasibility: All the technologies required in this are well established and their services can streamline different aspects like payment, shipping, authentication etc.

Resource Availability: There is a huge pool of suppliers in India (about **7 million artisans**) and also the department of posts is a reliable and cost efficient logistics network spread across **219 countries**.

Viability :-

Market Demand: There is a strong demand of authentic Indian products with assured quality by **73%** of the Indian diaspora which is over **32 million**.

Economic Viability: Indian artisans market is experiencing a significant growth (expected to **reach \$5.8 trillion by 2027**), utilizing it to cover development and operational cost through transaction cuts, advertisement etc. along with efficient logistics through DoP.

Regulatory compliance: Involves FDI regulations, GST registration and IT Act data protection. Security measures include encryption, multi-factor authentication and quality assurance ensuring data privacy and build trust.

Potential challenges and solution :-

- Challenge :** Product Quality Control
Solution: Implement a strict seller verification and review system, along with quality checks for all products before shipment to guarantee authenticity and reliability.
- Challenge :** Conflict Resolution in Product Customization
Solution: Use a legal written bond that contains all customization details. If the product doesn't meet the agreed terms, the buyer gets a refund otherwise no refund.
- Challenge :** Limited Digital Literacy Among Artisans
Solution: Provide training and support to artisans on using the platform, including easy-to-use interfaces and multilingual support to simplify the experience and adaptive process.
- Challenge :** Supply Chain Management
Solution: Implement real-time tracking and inventory management features to help artisans manage demand efficiently and ensure timely deliveries.

Benefits :-

- **Direct Access to Authentic Products** : Platform offer Indian diaspora and international buyers easy access to authentic products directly from local artisans and MSMEs in India.
- **Support for Local Artisans** : Empowering small-scale artisans by providing them access to the global market, **increasing** the **annual income** of artisans by **25-50%**
- **Standardized Shipping and Quality Control** : Guarantees fair, cheaper (**30-40% less**) & transparent shipping costs. Thorough quality check ensuring every product meets high standards and customer satisfaction.
- **Seamless User Experience** : Simple and convenient way for users to browse, purchase and receive traditional products globally, building a strong emotional connection with India.
- **Enhanced Trust and Transparency**: More than **60%** of Indian diaspora buyers report concerns about product authenticity and quality when purchasing Indian goods. The platform fosters trust through a transparent system of **seller verification, product quality checks, and reliable buyer feedback.**

Impact :-

- **Economic Impact**: Increases annual income of local artisans and MSMEs by **25-50%** and decreasing price to the customers by **30-50%** with the help of DOP.
- **Cultural Impact**: Preserves traditional crafts and heritage while elevating global awareness of Indian craftsmanship.
- **Technological Impact**: Revolutionizes e-commerce with AI and ML for personalized experiences, while delivering data-driven insights that empower sellers to optimize their products..
- **Global Impact** : Indian handicrafts exports are growing, with an rate **of 25.7%** year-over-year. Through our platform we will be Promoting India's cultural products globally improving the country's reputation for quality, diversity, and heritage, enhancing "Brand India" on the world stage.
- **Environmental Impact**: Promotes sustainable, handcrafted products. & Reduces reliance on mass-produced goods also reducing the usage of non ecofriendly materials.

- ❑ **Shipping cost comparison between DOP and private organization:** Indian Post VS FedEx
 - Indian Post: <https://www.indiapost.gov.in/VAS/Pages/calculatePostage.aspx>
 - FedEx: <https://www.fedex.com/en-us/shipping/international/understanding-international-shipping-cost.html>
- ❑ **Challenges faced by artisans:** [The Challenges Faced By Artisans In Developing Countries | Vceela - Online marketplace for handicrafts by artisans and artists](#)
- ❑ **Challenges and benefits of overseas shipping:** [The Benefits And Challenges Of Exporting Handmade Products Overseas || Rural Handmade-Redefine Supply to Build Sustainable Brands](#)
- ❑ **Artisans data analysis:** [Artisans in India and their quest for growth: Understanding the current set of capabilities among Indian artisans to adopt digital technology solutions – Catalyst Atal Incubation Centre \(catalystaic.org\)](#)
- ❑ **Diaspora Problem:** [Problem NRIs Face While Shopping from Indian Online Websites and How Forward Parcel Can Help - International Shipping from INDIA](#)
- ❑ **Handicrafts market size:** [Indian Handicrafts Market Size, Share, Analysis, Report 2024-2032 \(imarcgroup.com\)](#)
- ❑ **Global Data about diaspora:** <https://www.weforum.org/agenda/2019/09/indian-diaspora-largest-in-world-numbers/>
- ❑ **Indian handicrafts data:** <https://indian.handicrafts.gov.in/en>
- ❑ **Competitor analysis:** <https://ruralhandmade.com/> and <https://www.directcreate.com/shop>